

Yassine ABASSI

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EDUCATION

Bachelor of Engineering, Computer Engineering

Polytechnique Montréal

Aug 2022 – May 2026

Montreal, QC

EXPERIENCE

Software Quality Assurance Intern

UPTOTEST

May 2024 – Sep 2024

Ariana, Tunisia

- Designed and automated end-to-end functional test suites for the KOORS web app using Cypress with Cucumber/Gherkin in a BDD workflow.
- Translated product requirements into testable user stories and acceptance criteria, managed defects and traceability in Jira across sprints.
- Participated in Agile Scrum ceremonies to align QA priorities with development milestones and release timelines.

Private Tutor

Self-employed

Aug 2021 – Present

Montreal, QC

- Tutoring Mathematics, Physics, and Programming for college and university students.

PROJECTS

Hydro-Québec Weather Forecasting Platform | *Angular · .NET · EF Core · GDAL · Leaflet · SignalR · Docker* 2026

- Built a full-stack weather visualization platform with Angular and ASP.NET Core for multi-model weather display with interactive geospatial mapping and real-time updates via SignalR.
- Implemented GDAL raster processing and Leaflet cartographic rendering; designed RESTful APIs alongside an SQLite database, containerized the application with Docker and set up a CI/CD pipeline for automated testing and deployment.

3D Interactive Portfolio | *React.js · Three.js · Upstash Redis* | [Live Demo ↗](#) 2025 – 2026

- Crafted a 3D portfolio with an interactive room featuring a functional OS, custom camera rigging, and state-driven transitions across mobile and desktop, expanded the experience with terminal emulation, procedural art, and embedded mini games.
- Integrated a Gemini-powered chatbot assistant via a server function with Upstash Redis-backed rate limiting.

SSH Terminal Portfolio | *Go · Bubble Tea · Lip Gloss · Wish · Docker · Fly.io* | [Live Demo ↗](#) 2025 – 2026

- Built an SSH-based interactive terminal portfolio using Go and the Charm ecosystem (Wish, Bubble Tea, Lip Gloss), featuring animated splash screens, keyboard navigation, and dynamic theme switching.
- Deployed on Fly.io with Docker containerization, persistent SSH host key management via mounted volumes, and automated CI/CD through GitHub Actions.

Lost Woods Browser Horror Game | *React.js · Typescript · Web Audio API* | [Live Demo ↗](#) 2025 – 2026

- Built a canvas-based horror game with procedural forest tile generation, and reachable key placement using pathfinding logic.
- Engineered enemy AI behaviour, flashlight-reactive threat logic, stamina system, purge spell with cooldown, and separate indoor/outdoor scenes.

Multi-Robot Exploration System | *Python · ROS2 · Gazebo · Docker* 2025

- Designed a coordinated multi-robot exploration system with AgileX Limo robots.
- Implemented navigation and inter-robot coordination in Python while containerizing software modules using Docker.

Real-Time Multiplayer Combat Game | *Angular · NestJS · WebSockets* 2024

- Developed a real-time multiplayer game featuring seamless bidirectional communication.
- Designed event synchronization and matchmaking logic to ensure smooth gameplay under concurrent loads.

Embedded Obstacle Detection System | *C++ · ATmega164A* 2023

- Built an autonomous obstacle detection and search system with low-level I/O control, and real-time decision logic on the ATmega164A microcontroller.

Desktop Chess Application | *C++ · Qt* | [Github ↗](#) 2022

- Built a fully interactive chess application implementing legal move validation with a custom QWidget board renderer.

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, Go, Kotlin, Swift, VHDL, Assembly

Web & Mobile: Angular, React.js, Node.js, NestJS, Express.js, WebSockets, SignalR, .NET, SwiftUI, Jetpack Compose, Three.js

Data & APIs: PostgreSQL, MongoDB, SQLite, Firebase, Retrofit, OkHttp, Entity Framework Core, GDAL, RxJS, Upstash Redis

DevOps & Tools: Git, GitLab, Docker, Jira, Netlify, Fly.io, Swagger, Qt, Cypress, Jest, Jasmine, Mocha, Vivado, Wireshark

Robots & Boards: Zynq7000, ATmega164A, Pynq-Z1, Basys3 Artix-7, AgileX Limo

ADDITIONAL

Hackathon: PolyHacks 2025 - Built **WildGuard**, an AI-powered wildlife monitoring system that detects and classifies animals in real-time using YOLOv8, data pipelines and a Flask web interface for live video streaming and alerts. [Github ↗](#)

Languages: English, French, Arabic (fluent) | German (intermediate)

Work Authorization: Authorized to work in the U.S. and Canada (dual citizen)